

Advances in design and manufacturing of environmentally friendly and biocide-free antifouling/foul-release coatings: replacement of fluorinate species

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Abstract Amphiphilic coatings are nowadays described as the most promising strategy for the achievement of an effective and eco-friendly antifouling marine paint, based on a physical principle. Nevertheless, most of the preparations imply the introduction of fluorinate chemicals for the formation of the hydrophobic moieties, which are potentially toxic for the marine environment in the case of an accidental release. The present work evaluates the possibility of removing fluorinate species for the implementation of such coatings and their replacement with analogous long-alkyls. In particular, this work presents an easily implementable procedure for the manufacturing of an amphiphilic system that exploits the reactivity of a commercially available –OH-rich hybrid polysiloxane resin. The resin was first modified through sol-gel chemistry that was used in the grafting of a silane-bearing hydrophobic long-alkyl chain, while the introduction of the hydrophilic moieties was conducted contextually to the curing process, through the reaction with a commercially available PEG-olate isocyanate hardener. The effective persistence of the amphiphilic character with respect to the replacement of the fluorinate silane by the alkylsilane was ensured by the measurement of the hysteresis contact angle. The effectiveness of the persistence of the antifouling/foul-

release properties was assessed through testing against the adhesion and deposition of egg white, as a proteinaceous probe, *Mytilus edulis* (mussel) and *Ulva intestinalis* (algae). All the tests indicated that the amphiphilic coating prepared from long-alkyl silane performed even better than fluorinate silane-containing samples in this hybrid coating system resulting from the fusion of a xerogel and a resin. The results suggested that the use of a long alkyl silane was an effective alternative to the use of potentially noxious fluorinate silanes, with advantages that boost the ecological benefits and also include the performance of the coating.

Keywords Antifouling, Fouling-release, Long-alkyl chain, Amphiphilic, Coating

Introduction

The application of an efficient antifouling paint is fundamental for fuel saving, emissions, and the costs related to the cleaning and maintenance cycles of boats.¹ So far, many strategies have been evaluated to replace the traditional antifouling paints based on noxious biocides.² Many attempts have been conducted to achieve a biocidal effect dispersing active safe chemical substances within a traditional matrix. The photocatalytic activity of ZnO and TiO₂ nanoparticles was exploited to modify coatings able to destroy biofilms by an accelerated oxidative process.^{3–5} This effect is, however, strongly dependant on UV exposure and, consequently, is not equally effective on all the parts of the hull.

Low-surface energy coatings⁶ are based on polydimethylsiloxane (PDMS) and derivatives^{7–9} or fluorinate polymers^{7,10} allowing the establishment of only weak bonds with marine biofilms. Similarly, the establishment of weak bonds can be achieved through the

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implementation of surfaces featuring a very dense structure at a molecular level.¹¹ Saravanan et al. reduced the porosity of a glycidyl carbamate-based epoxy resin turning it into a hybrid PU–epoxy resin by the addition of an isocyanate hardener. Highly hydrophilic surfaces are thermodynamically more favorable to the soaking of the surface rather than to the deposition of proteinaceous substances.^{12,13}

In this framework, the more effective amphiphilic coating systems join together the antifouling/foul-release (AF/FR) properties of the former two strategies¹⁴ alternating hydrophilic and hydrophobic domains in a fine patterning. Moreover, this arrangement can distort the globular structure of the gluing proteins secreted by marine organisms exploiting the natural adaptability to the polarity of the surface of adhesion.¹⁵

The chemical mosaic can be implemented through the addition of nanoparticles characterized by an amphiphilic surface dispersed in a base resin. Examples are the coating systems elaborated¹⁶ where nanoparticles of hydrophobic polyhedraloligomeric silsesquioxane were modified by the presence of hydrophilic NH₂. However, this implies the use and the release into the environment of potentially noxious nanostructures. Amphiphilic surfaces can be more often obtained from pure chemical mosaics, without the addition of any 3D structure. This implies several reaction pathways, including atom-transfer radical polymerization,¹⁷ conventional esterification reaction,¹⁸ free radical copolymerization,¹⁹ or a more easily implementable sol–gel reaction. This last reaction pathway provides the condensation of functionalized silanes on –OH groups eventually available on a substrate.^{17,20–26} Many studies have been conducted to extrapolate the effect each functional grafting group has on the AF/FR properties of an amphiphilic or low-adhesion coating. The influence of the length of alkyl chains of thiols species used for the implementation of SAM foul-release coatings was studied by Bowen et al.²⁷ The authors showed how the increase in the number of carbon atoms over 12 not only led to the decrease of frictional forces (μ) but also determined the arrangement of the lateral chains in a tidy configuration that is energetically unfavorable to the deposition of marine organisms. Wang and Zuilhof²⁸ compared the AF/FR properties of fluorinate components arranged in diverse configurations: homo-polymers, co-polymers and superstructures, in brush systems. The highest AF/FR character was found with homo-polymers since they allow a tightly packed arrangement. The configuration of the lateral chain was also evaluated studying ternary styrene co-polymer systems grafted with short fluoroalkyls or perfluoropolyethers.²⁹ The effectiveness of the AF/FR properties was found to be related to the structure of the lateral chain which indirectly influences, at a given fluorinate monomer content, the capability of the hydrophobic moiety to migrate on the outmost surface. An indirect comparison between the use of perfluorinated alkyl silanes and alkyl silane was conducted to evaluate the efficiency of a xerogel release coating

against the fouling operated by *Cellulophaga lytica*, *Navicula incerta*, and *Amphibalanus amphitrite*.²³ The AF/FR properties were found to be proportional to the average surface energy of the coating obtained by blending alkyl fluoro silanes and alkyl silanes of different polarity regardless the overall content of fluorine of the coating. Moreover, the study showed that, at a given polarity, roughness, and elastic modulus of the surface, AF/FR properties were strongly determined by the capability of the active functionalities to segregate and arrange in a chemical mosaic on the coating surface. This tendency was found to be more pronounced with fluorinate reagents. Similarly, Tang et al.²⁵ demonstrated that the AF/FR properties vary strongly with the surface energy, following a relationship that depends on the contaminating species. This was the main determining factor in the tested series of release coatings obtained from the blending of different proportions of short and long, fluorinate, alkyl silanes.¹⁵ The capability of a surface to achieve a low surface energy is mainly related to the tendency of the migrating of the hydrophobic moiety rather than to its absolute content of hydrophobic functionalities.²⁹ Since the integration of an efficient AF/FR hydrophobic domain in an amphiphilic surface is able to improve the overall AF/FR properties of the surface, fluorinate silanes are, in this sense, the most promising functionalities. Their critical energy, the lowest among other possible candidates according to the Bayer curve, would endow the fluorinate species with the highest tendency to segregate from most parts of the resins adopted for the manufacturing of paints.¹⁰ Nevertheless, even if fluorinate silanes appear to be the ideal choice for the implementation of the most effective family of coatings, it entails public concern due to the persistence of the noxious substances in the environment as a result of the stability of the C–F chemical bond. This stability results in the incapacity of many species to metabolize these chemicals that bio-accumulate in the serum and tissues of animals and vegetation, causing acute and chronic toxicity in liver tissues (hypertrophy), the thyroid, and the blood cell wall, and incorrect absorption of triglycerides in humans and other mammals.³⁰ For this reason, the voluntary complete removal of fluorinate substances from the manufacturing of painting products would be advisable, in particular inside antifouling marine paints, subjected to harsh environmental conditions and direct contact with water.

This study will describe the implementation of two different amphiphilic coatings using 1H,1H,2H,2H-perfluoro octyltriethoxysilane (FC8-Si) or hexadecyltrimethoxysilane (C16-Si) as alternative to fluorinate composites for the constitution of the hydrophobic phase of an AF/FR amphiphilic system based on the fusion of an organic–inorganic polysiloxane resin and xerogel under the form of grafted silanes. The functionalized silanols will graft to –OH groups of a hybrid organic-polysiloxane resin through hydrolysis and condensation reactions, selected as binder of the

paint system. A correlated amount of hydrophilic moieties will be introduced under the form of PEG-ilate isocyanate hardener, with the double purpose of curing and functionalizing the coating. The so-obtained systems have a great potential of technological transfer since they are obtained from commercially available products, and, moreover, they are based on corrosion resistant, mechanically robust, binder resin and hardener. The effect of the two hydrophobic silanes on the parameters governing the efficiency of the antifouling properties will be evaluated and related to the overall efficiency of each coating. Experimentally, the intrinsic hydrophobicity of each silane, the spatial arrangement of lateral chains, and the geometry of their patterning on the coating surface will be investigated through scanning electron microscopy (SEM) observations and measurement of the wetting properties. The variations in these parameters will be then correlated with the AF/FR efficiency of the coatings through testing executed using a proteinaceous probe (namely egg white), as well as real marine organisms on a laboratory scale. To the best of our knowledge, no paper reported a direct comparison between the efficiency of AF/FR coatings, obtained through the sole use of long alkyl or fluoroalkyl silanes, in an amphiphilic coating system based on the use of binder resin joined to a silane/xerogel system.

Experimental

Materials

The hydrolysis reaction of the fluorine-containing sample was conducted pouring the fluorinate silane (Dynasylan F8261; Evonik, Essen, Germany) in a vessel containing HCl (Sigma Aldrich, Milano, Italy) as hydrolysis catalyst, acetone (Sigma Aldrich) as reaction medium, distilled water, and tetraethyl orthosilicate (TEOS) as compatibilizer (Sigma Aldrich). The adopted silane was a C8 perfluorinated silane, 1H,1H,2H,2H-perfluorooctyltriethoxysilane. The solution was reacted under vigorous magnetic stirring for 2 h at room temperature. A fixed aliquot of an -OH-rich hybrid polyester-polysiloxane (Silikotop E901; Evonik) was then added to the hydrolysis mixture for 1 h for the grafting reaction under identical conditions. The hydrolysis step of C16 alkyl silane, hexadecyltrimethoxysilane commercialized under the name of Dynasylan 9116 (Evonik), was conducted contextually to the grafting step, dissolving 0.2 g of diethanolamine (Sigma Aldrich) as a catalyzer in 67.5 ml acetone (Sigma Aldrich), 10.5 ml distilled water, and an aliquot of 13.65 ml of Silikotop E 901 to obtain the grafted resin used in the preparation of sample Alkyl2. The mixture was reacted with magnetic stirring for 2 h at room temperature.

Samples obtained from FC8-Si and evaluated during the preliminary steps were prepared according to the

quantities reported in Table 1. These quantities are estimated for a content of resin of 1.81 g, a volume of acetone of approximately 8 ml (the volume required to achieve a final volume of 10 ml), and a volume of hydrolysis catalyst (HCl) of 43 μ l. The optimized proportions, described later (see “[Assessment of amphiphilic coating formulation](#)” Section), will be indicated as sample F1.

The introduction of the hydrophilic moieties was contextual to the curing step, being that these were constituted by a PEG-ilate isocyanate hardener (Bayhydur 3100; Bayer-Covestro, Leverkusen, Germany). Antifouling coatings were spray-applied on an epoxy primerizing plaster (Veneziani Yachting, Boero, Genova, Italy), applied in turn on stainless steel slabs (AISI 304). A white one-pot non-anti-vegetative top coat (Unigloss; Veneziani Yachting) was applied directly on the same plaster to obtain reference samples (Ref. 3). All the samples were cured at room temperature using dibutyltin dilaurate (DBTL) as a catalyst at a concentration of 1 wt% calculated on the resin amount. The catalyst, even if an organic composite of tin, does not have the same harmful effect on marine organisms in the environment as tin composites deliberately designed as biocides. Some tests reported in the Appendix can be adduced to corroborate this statement (see “Additional Material”).

Evaluation of reaction effectiveness

The effectiveness of the grafting process of the resin by means of alkyl silane was verified by attenuated total reflection Fourier-transform infrared (FTIR) spectroscopy. For comparison, a grafting solution without the addition of any catalyst, HCl, an aliquot of the as-is resin, and an aliquot of the as-is silane were also examined. A spectrophotometer (Jasco 6600; Jasco, Easton, MD, USA) was used for this purpose, analyzing the acquired spectra with the help of the KnowItAll database (Bio-Rad, USA). All possible interfering signals related to the presence of the solvent, unreacted monomers, or HCl residuals were avoided by performing a vacuum treatment. For this purpose, all samples were dried overnight in a vacuum oven (M40-VT; MPM Instruments, Bernareggio, Italy) since the interfering substances are very volatile.

Analysis of amphiphilic character

The amphiphilic character of the samples was quantified through the determination of the wetting hysteresis, calculated as the difference between the advancing and the receding contact angles of a water drop. A 100- μ l drop of distilled water was poured through a square-cut hollow needle of a syringe. The piston of the syringe was attached to and moved by a motorized cross-head with a speed of 8 mm/min. A photograph of the drop profile was taken every 0.5 mm of displace-

Table 1: Composition of samples evaluated during the preliminary testing

Sample	H ₂ O (μl)	TEOS (g)	FC8-Si	Hardener (g)
A	380	0.81	0.75 g	0.23
B	100	0	0.75 g	0.23
C	100	0	0.75 g	0.46
D	100	0	0.75 g	0.70
E	100	0	0.75 g	1.40
F	380	0.81	0.75 g	0.46
G	380	0.81	0.75 g	0.70
H	380	0.81	0.75 g	1.40
I	380	0.81	0.75 g	2.8
L	380	0.81	0.75 g (ratio F:PEG = 1:3,4)	6.4
M	380	0.81	Ratio 1:100 = 32 μl Ratio 1:20 = 160 μl Ratio 1:15 = 216 μl Ratio 1:10 = 320 μl Ratio 1:1 = 3.2 ml	6.4

ment of the piston during the expulsion phase (10 μl of liquid ejected), and every 0.25 mm (0.5 μl of liquid) during the aspiration. Images were acquired using a Canon D60 macro lens (100F28; Canon Italia, Milano, Italy), and elaborated by means of the AutoCAD graphics program. The measurements were repeated in four replicates for each sample. Surface morphology, and the possible presence of segregation phenomena at the microscopic level, were observed using an emission gun SEM (LEO Supra 35) set at 10 kV. Images were acquired using detectors for back scattered electrons (RBSD) and secondary electrons (SE2).

Antifouling evaluation: egg white test

Egg white was selected as the testing probe to evaluate the overall antifouling properties of the experimental samples. The test was conducted by pouring 1 ml of egg white on the surface of the sample under examination and left to dry overnight.

The substance, composed of a 50 wt% of water and 50 wt% of albumin and other glycoproteins.³¹ OVO-albumin (OVA) was selected because of the similarity of its components to bovine serum albumin (BSA). Both these globular proteins are, indeed, amphiphilic and soluble in water³² thanks to the easy exposition of hydrophilic amino acids on their external surface.³³ Moreover, the two albumins share a similar amino acidic sequence³⁴ responsible for the similar behavior in the adhesion mechanism of the two molecules on different substrates, even when these are extremely peculiar, such as specific antibodies.³⁵ BSA is largely used as a testing probe in the preliminary evaluation of the overall AF/FR properties of a functionalized coating.^{36–38} Moreover, the analogies occurring among OVA, BSA³⁹ and gluing substances of marine organ-

Table 2: Criteria adopted for the assignation of marking scores to egg white test results

Grade mark	Fouling entity
0	Clean surface after shaking
1	Clean surface after brushing
2	20–70% of egg residuals left on the surface
3	70% < of egg residuals left on the surface

isms also include the proteins contained in the egg white. These feature an amphiphilic behavior and can be adopted as a suitable testing probe, as reported in Refs. 40 and 41.

The arrangement of the dried probe gave an immediate occurrence of the adhesion strength of the substance to the surface. Table 2 illustrates the ranking criteria.

Antifouling evaluation against marine organisms

The repulsion of marine organisms was evaluated by the immersion of three replicates of each formulation using a different triplet for each test, in a 15-l aquarium filled with natural marine water, an amount of about 300 g of wet algae (*Ulva Intestinalis*), and *Mytilus edulis* (mussels) collected on the beach of Ostia (Rome, Italy) (Figs. 1a and 1b).

A first test was conducted to evaluate the repellency of the surface to algae. The extent of the fouling was calculated as the percentage of surface coverage of the algae after 15 days of immersion in the aquarium and 20 s of flushing with running water to simulate the shear stresses applied during the navigation. The coverage was determined through a colorimetric method, exploiting the differences between the green

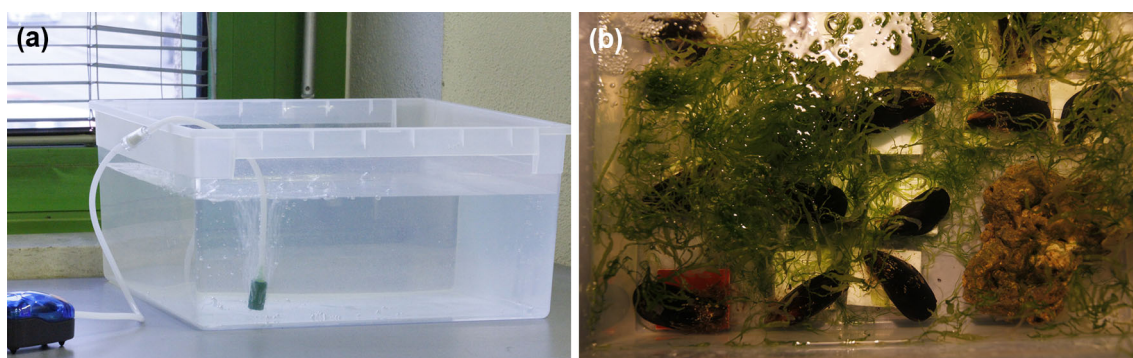


Fig. 1: Preparation of aquarium system: (a) oxygenation system; (b) disposition of *Mytilus edulis* on the samples

Table 3: Evaluation scale of fouling due to *Mytilus edulis*

Grade mark	Fouling entity
0	Byssus filament easily removed by a gentle rubbing with a cotton bud
1	No live mussel on the sample, no byssus filaments attached
2	Mussel remains on the sample without byssus filaments attached
3	From 1 to 4 byssus filaments attached
4	More than 4 byssus filaments attached

algae and the white surface and expressed as the average percentage of the covered surface. ImageJ bundled with 64-bit Java 1.8.0_112, was adopted as the colorimetric software. The test measurements were conducted on an area of 3×4 cm.

The evaluation of AF/FR properties against molluscs was studied through a procedure previously adopted in the relevant literature with analogous purposes.⁴² Two molluscs were placed on the tile surface, with the first of the two oriented with the foot facing the internal side of the tile, while the second was placed oriented in the opposite direction. The mussels were not physically immobilized onto the tile to allow them the possibility to select an adhesion surface according to their preference. Indeed, Gittens et al.⁴³ reported an evaluation step of a potential substrate prior to the ultimate anchoring, with a number of adhesion plaques produced that were proportional to the suitability of the chosen surface. This last parameter was included as a criterion for the evaluation of the AF/FR performance of the tiles after 4 days of immersion. The score mark was assigned following the description in Table 3. Adhesion strength was also considered for the determination of the score mark.

The highest mark was assigned to the scenario where the byssus was easily removable by gentle rubbing on tile surface using a cotton bud. The rubbing simulates the foul release action of the paint during navigation at low speed (less than 8 nodes). This was considered the most advantageous behavior since the adhesion properties of plaque are recognized as one of the strongest achievable in nature. For this reason, the differences in the AF/FR properties of the diverse coatings can only be distinguished through the number

of filaments attached, with the only exception of extremely fouled release surfaces. On these surfaces, the attached mussels can be easily removed under the action of a cotton bud applying negligible shear stresses compared to the ones occurring during navigation.

Results and discussion

Assessment of amphiphilic coating formulation

In this section, the results obtained studying different hydrophilic and hydrophobic contents are reported. The assay was originally designed to have 1/3 of all the available $-OH$ groups on the resin covalently bound to a hydrophobic silane, 1/3 of the groups available to react with hydrophilic isocyanate hardener, and 1/3 free $-OH$ groups left unreacted to give the system a higher flexibility.

This content of functional lateral chains shields to a larger extent the intermediate polarity of the polysiloxane resin (Figs. 4 and 5). A reaction scheme of the process, whose effectiveness was verified by FTIR as reported in “FTIR analysis of the effectiveness of grafting process” Section, is reported in Figs. 2, 3, 4 and 5. Figure 2a shows the hydrolysis reaction of fluorinate silane, Fig. 2b the hydrolysis reaction of alkyl silane, and Fig. 2c the hydrolysis of the compatibilizing agent, TEOS.

The “hydrolysis step” activates the hydrophobic silanes through the removal of alkoxy groups letting the free $-OH$ groups, which will be involved (Fig. 3) in

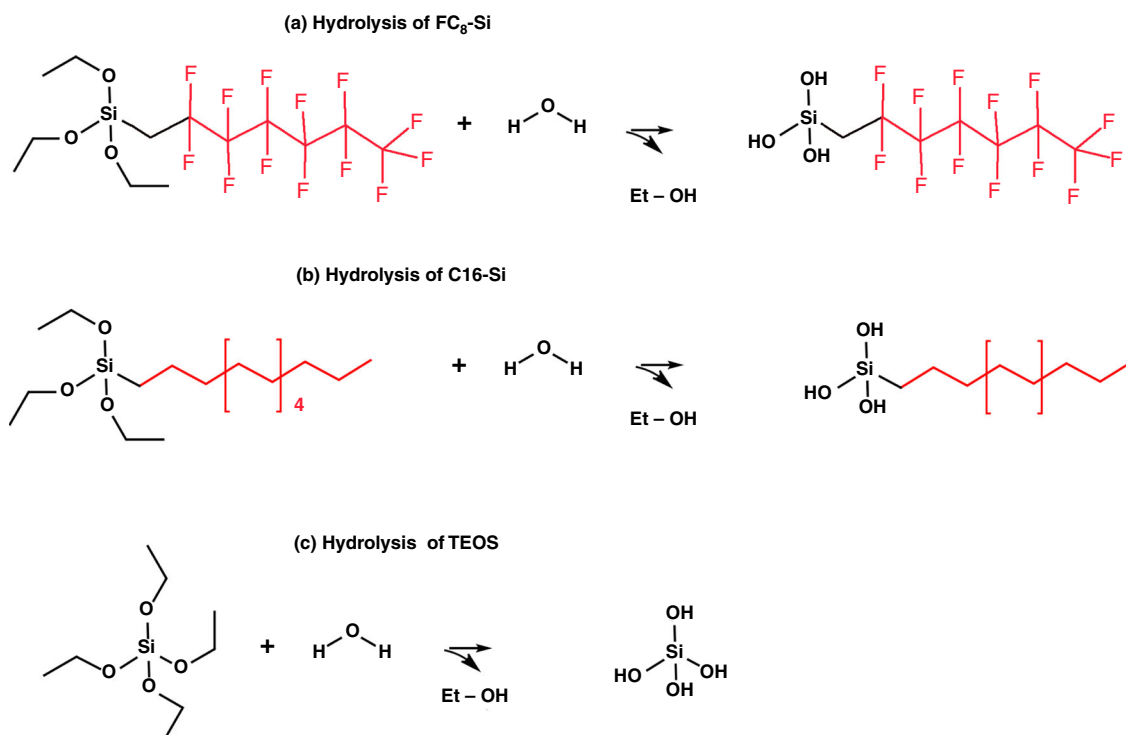


Fig. 2: Hydrolysis reaction for the activation of (a) fluorinate hydrophobic silane, (b) hydrophobic long alkyl silane, (c) compatibilizing agent, TEOS

the “grafting step”, occur in a condensation reaction with the -OH groups on the resin and the elimination of a water molecule.

Due to the high reactivity of alkyl silane, manifested in preliminary synthesis attempts with the instantaneous formation of a white precipitate, the hydrolysis and the grafting steps of this last reactant were carried out in one single step adopting a slower alkaline catalyst, namely diethanolamine.

Tests results are described in Table 4 for each combination of hydrophobic moieties, compatibilizing agent, hydrophilic hardener, and hydrophilic/hydrophobic ratios. The exact experimental quantities of reactant are reported in Table 1.

The first trait emerging from the experimental campaign was the very low compatibility of hydrophobic FC8-Si with the remaining components of the system. Initial attempts to conduct the hydrolysis step without the addition of any solvent, in the presence or absence of TEOS as compatibilizer, led to the almost instantaneous formation of a white precipitate (Fig. 6), as a symptom of a phase separation (results not reported).

The introduction of a medium polarity solvent allowed the correct conduct of the first two steps of the paint manufacturing process, namely hydrolysis and grafting, being that all the species were soluble in it. Moreover, the presence of TEOS in preparations F, G, H, I, L, and M was essential to avoid the clouding

phenomenon given by the phase separation in solutions B, C, D, and E.

The curing step, initially planned to be carried out using a normal concentration of trifunctional isocyanate corresponding to the CNO:OH ratio of 2/6 (C, F), was unsuitable for the two systems analyzed, whether or not the presence of TEOS was implied. A similar result was logically found for samples A and B prepared using lower concentrations of hardener (CNO:OH ratio of 1/6). Nevertheless, the increase in isocyanate content up to a ratio of 3/6 (D and G) was also unsuitable to give place to a properly dried coating. A suitable content of hardener was finally worked out for samples that did not contain TEOS (E) with an increase of hardener concentration up to a CNO:OH ratio of 6/6. Yet, solution H, exploiting the compatibilizing properties of TEOS, persisted in an uncured state even after the addition of the same amount of hardener as for solution E. The doubling of this amount of hardener did not achieve any improvement in the cured state of formulation I, a preparation containing TEOS. These last observations suggested that the introduction of an extra amount of -OH groups, derived from the grafting reaction (see Fig. 3), interfered with the curing process, reacting with isocyanate groups initially intended for the direct reaction with -OH groups on the resin. This last hypothesis was confirmed by the proper cured state of sample L and most of the samples from the M group as

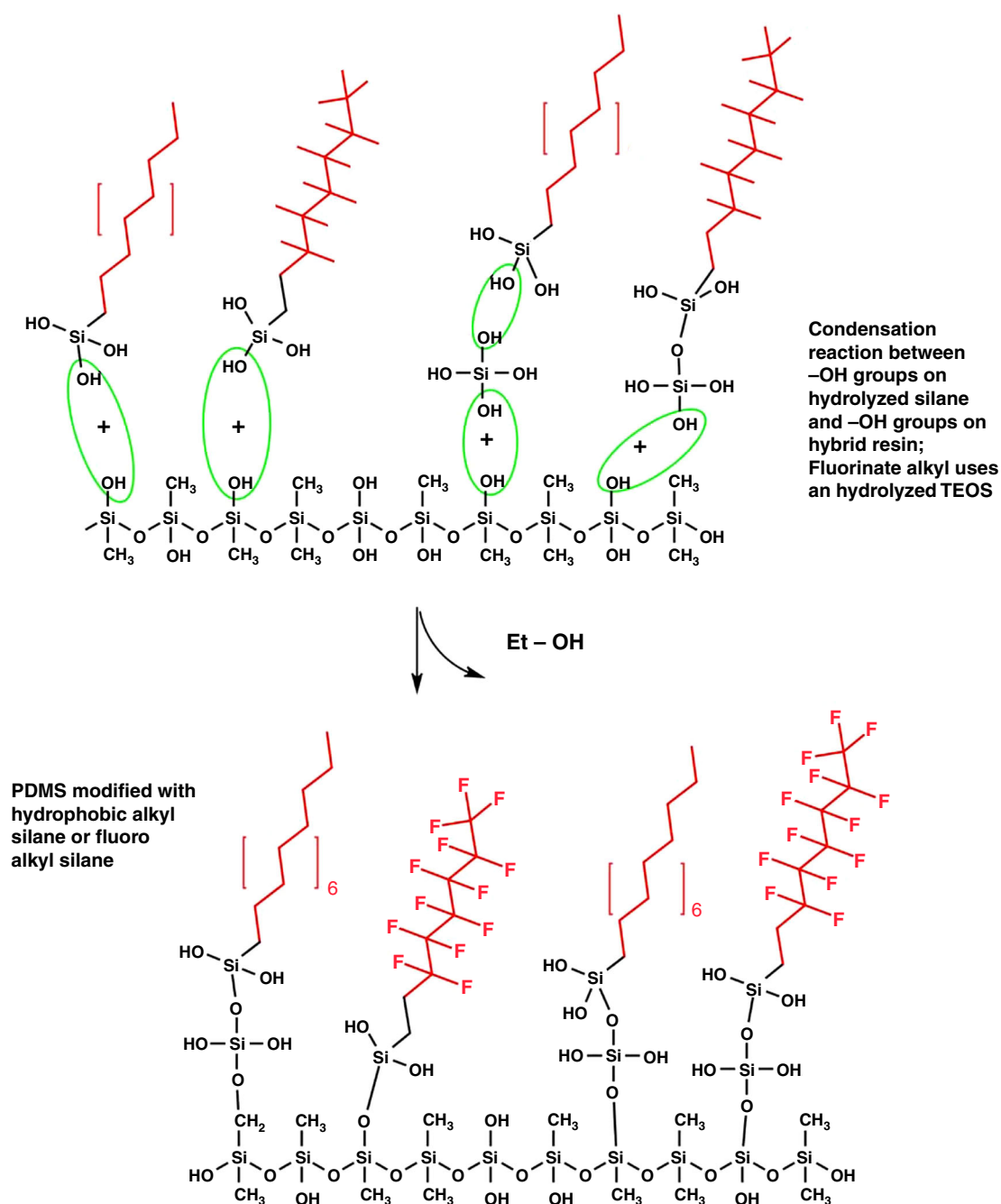


Fig. 3: Scheme of the grafting reaction occurring with hydrophobic silanes

they received an extra amount of hardener so as to have a CNO:OH ratio of 6/6. Indeed, in this case, the number of $-OH$ was calculated also taking into account the functional groups introduced through the addition of TEOS (4 $-OH$ groups for each TEOS molecule).

The finishing of the surface required a further adjustment, since it showed a visible segregation phenomenon among the components (Fig. 7). With a fixed content of PEG-ylate hardener, many concentrations of fluorinate silane were tested in order to obtain several different F/PEG ratios, or several different

absolute contents of fluorinate silane. The evaluated ratios, ranging from 1:1 to 1:100, are indicated as solution L and solutions of group M. It is evident from the images (Figs. 7a and 7b) that concentrations of FC8-Si higher than the ones corresponding to the ratio 1:10 of group M led to a segregation phenomenon during the curing step of the coating. This phenomenon occurred despite the persistent transparency of the reacted resin even many days after the reaction, indicative of a perfect compatibility and the correct reaction between FC8-Si and the resin. This observa-

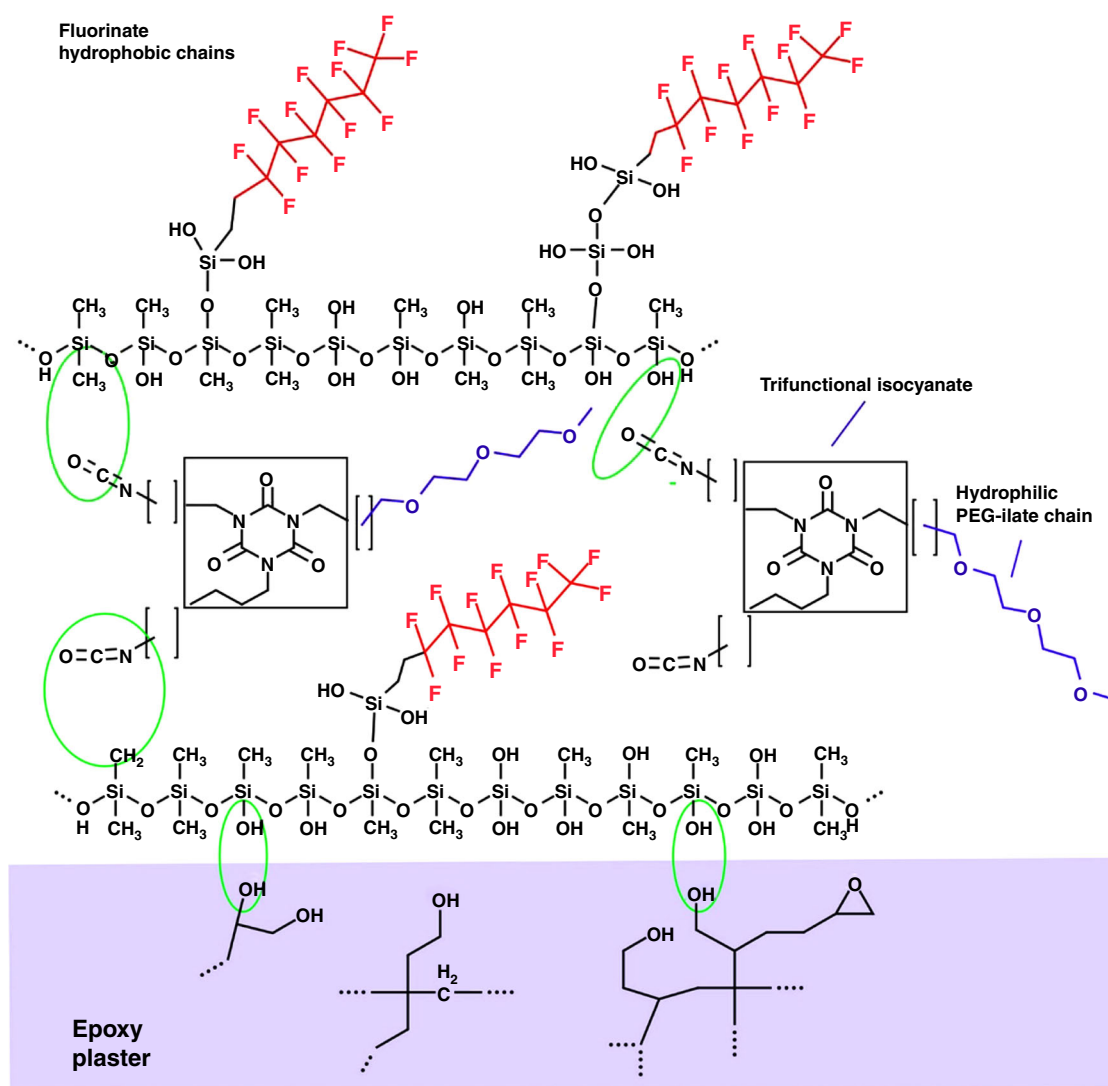


Fig. 4: Scheme of the curing reaction occurring between -OH groups on resin and hydrophilic isocyanate, sample F1

tion led to the conclusion that the separation was mainly due to the repulsion between the modified resin and the highly hydrophobic isocyanate. The ratio 1:1 did not reach a proper curing due to the excessive overall content of silanes.

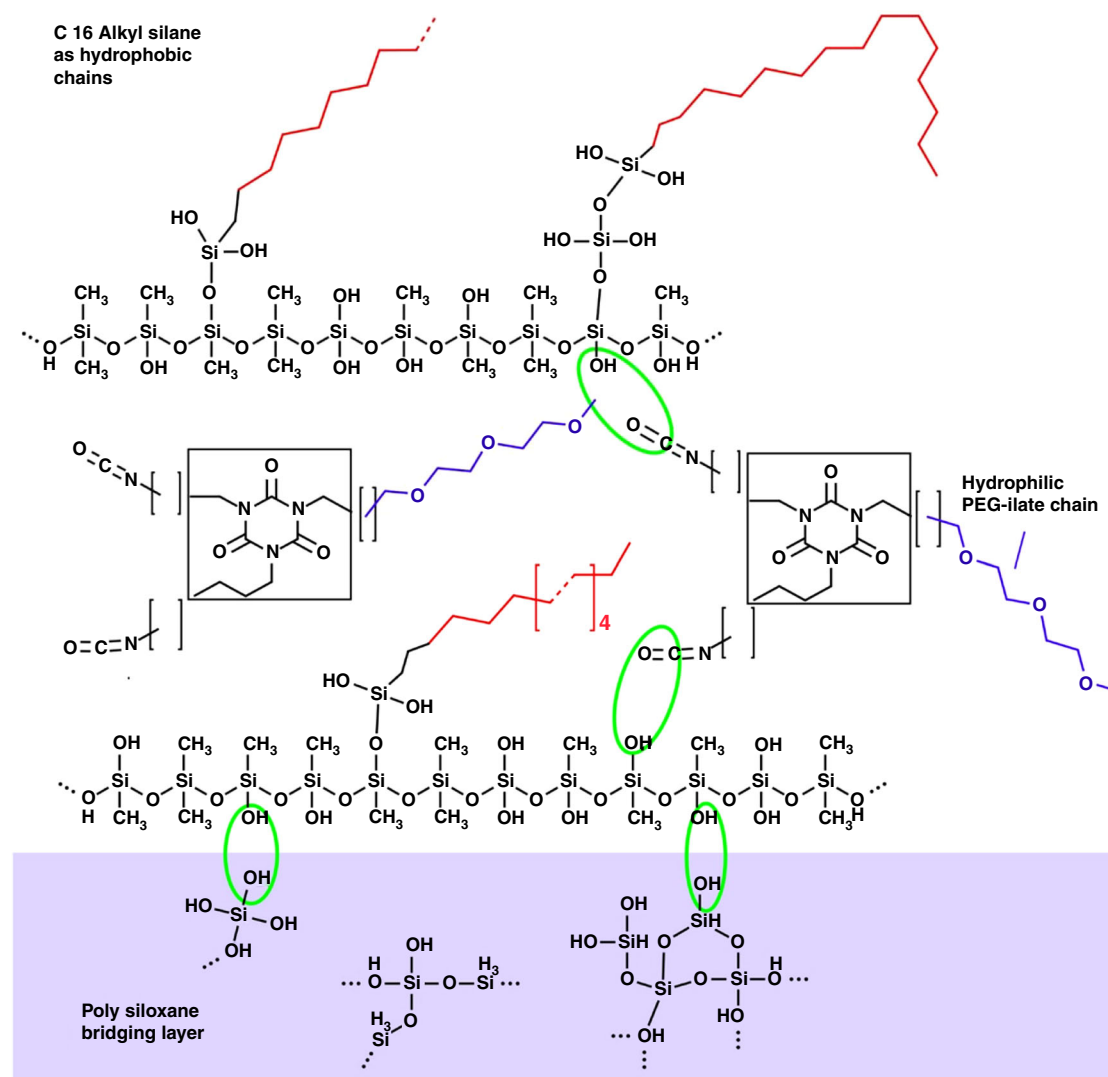
The smooth finishing of samples obtained through the adoption of F/PEG ratios included between 1:15 and 1:100 confirmed the hypothesis (Fig. 7c). In this framework, the formulation that more closely reflects the coating composition originally proposed is the sample belonging to group M featuring a F/PEG ratio of 1:15. This sample will be indicated as F1 hereafter.

The higher critical energy featured by C16-Si allowed the implementation of an amphiphilic coating following a composition that more closely reflects the ideal formulation initially proposed. Indeed, a normal concentration of C16-Si corresponding to 1/3 of all the -OH groups available on the resin was easily added without the occurrence of any visible segregation

phenomenon. The CNO:OH ratio required to achieve a proper curing was 6/6. Even if this value is not equal to the one originally designed (CNO:OH ratio equal to 1/3), probably due to intrinsic steric features of this molecule, it still gets closer to this value than in the F1 sample. The lower amount of necessary hardener is in accordance with the absence of TEOS in this formulation, as a consequence of the higher polarity of C16-Si.

FTIR analysis of the effectiveness of grafting process

To evaluate the efficiency of grafting reactions, the FTIR spectrum of unreacted mixtures of each silane (indicated by UR_AlkylSi-Silik in the case of the introduction of alkyl silane) and resin was compared with the spectrum obtained on the processed resin



(indicated as AlkylSi-Silik in the case of grafting with alkyl silane). Analysis results of grafting processes using the fluorinate silane have been reported elsewhere.⁴¹ Grafting with fluorinate silanes presented many analogies with the results of the analysis on the grafting process conducted with alkyl silanes.

The absorption peak occurring at 3476 cm^{-1} in UR_AlkylSi-Silik and in the spectra of the as-is resin was related to the stretching vibration of -OH groups on the main backbone of the resin, as reported in Refs. 44–46. The same absorption shifted to a wavenumber of 3400 cm^{-1} after the grafting process in Alkyl-Silik (Fig. 2a) featuring, at the same time, an increment of the intensity. These two aspects represent evidence of the effective involvement of -OH groups on the resin in the grafting process, as expected by the reaction pattern described in Fig. 4. The shift of the peak towards a lower wavenumber can be justified by the appearance of -OH groups deriving from the hydrol-

Table 4: Results obtained during the optimization of experimental formulations

Sample	Main features	Curing	Finishing	Reactivity
A	1/3 OH involved in grafting reaction with silane CNO:OH ratio = 1/6 TEOS	NO	–	Clear and stable solution after the grafting reaction
B	1/3 OH involved in grafting reaction with silane CNO:OH ratio = 1/6	NO	–	Clear solution not stable over time
C	1/3 OH involved in grafting reaction with silane CNO:OH ratio = 2/6	NO	–	Clear solution not stable over time
D	1/3 OH involved in grafting with silane CNO:OH ratio = 3/6	NO	–	Clear solution not stable over time
E	1/3 OH involved in grafting reaction with silane CNO:OH ratio = 6/6	OK	NO: phase separation at later stages	The lack of TEOS allowed the curing of the coating with a OH/CNO ratio of 1:1
F	1/3 OH involved in grafting reaction with silane CNO:OH ratio = 2/6 TEOS	NO	–	Compatibilization achieved through the simultaneous use of TEOS and acetone
G	1/3 OH involved in grafting reaction with silane CNO:OH ratio = 3/6 TEOS	NO	–	
H	1/3 OH involved in grafting reaction with silane CNO:OH ratio = 6/6 TEOS	NO	–	
I	1/3 OH involved in grafting reaction with silane CNO:OH ratio = 12/6 TEOS	NO	–	CNO groups exceed OH functionalities on the resin. The cured state is not achieved since the this amount of hardener does not take into account the amount of OH coming from the silane (TEOS + FC8-Si) condensation
L	1/3 OH involved in grafting reaction with silane CNO:OH ratio = 12/6 + 1/(4x moli TEOS) TEOS	OK	Faint signs of segregation on cured coating	CNO groups are calculated by adding an extra amount of CNO equal to the number of –OH potentially coming from unreacted TEOS (worst-case scenario). This configuration resulted in a F/PEG ratio of 1:3,4
M	Variable number of OH involved in grafting reaction with silane CNO:OH ratio = 12/6 + 1/(4x moli TEOS) TEOS	ratio 1:100 = OK ratio 1:20 = OK ratio 1:15 = OK ratio 1:10 = OK ratio 1:1 = NO	ratio 1:100 = OK ratio 1:20 = OK ratio 1:15 = OK ratio 1:10 = irregular finishing	After having set the HD concentration required for the curing, TEOS concentration to have a homogeneous coating, the amount of silane was changed to give several different F/PEG concentrations. Despite the presence of TEOS, samples with the highest F/PEG ratios gave a visible surficial segregation

silanes and the as-is resin. In particular, peaks from the as-is silane at 2957, 2928 and 2852 cm^{-1} and peaks from the as-is resin at 2957 and 2877 cm^{-1} are all found in the Alkyl-Silik spectrum (Fig. 8a).

The increase in the absorption intensity of peak at 1043 cm^{-1} and its related shoulder at 1028 cm^{-1} confirmed the occurrence of a grafting reaction through a proper hydrolysis and condensation reaction (Fig. 8d). These two peaks arise from the asymmetric vibration of Si–O–Si bonds that constitute the main backbone of the as-is resin, and from the Si–O–Si structure obtained as the product of the condensation reaction between two molecules of functionalized silane and/or between the silanes and the resin.^{47,48} At the same time, the decrease of peak intensity at 1080 cm^{-1} in Alkyl-Silik with respect to UR_Akyl-Silik and the spectrum of the as-is resin is indicative of the effectiveness of the hydrolysis reaction (Fig. 8d).

As reported in Ref. 41, the occurrence of some grafted products deriving from minor alternative side

reactions was assessed by the study of the carbonyl signals. Indeed, an absorption peak appeared at 1650 cm^{-1} in the Alkyl-Silik spectrum as a peak shoulder of the main carbonyl peak occurring at 1730 cm^{-1} in all the analyzed spectra (Fig. 8b). The frequency and the intensity of this minor absorption was compatible with some nucleophilic addition occurring between OH groups on hydrolyzed Alkyl-Silik and the carboxyl groups on the polyester moiety of the hybrid resin with the consequent release of some of the polyester part. The occurrence of this kind of nucleophilic addition was corroborated by the decrease of absorption peaks at 1594 and 1430 cm^{-1} (Fig. 8c). Both these absorptions are, indeed, compatible with vibrations of C=C/C–C on aryl groups⁴⁹ that would be eventually removed in the case of nucleophilic addition on carbonyl groups of the resin. Aromatic functionalities are very likely to be present on the polyester fraction of the binder resin,⁵⁰ as also indicated by the presence of an absorption peak at 1457 cm^{-1} (Fig. 8c) and an absorption shoulder at 1110 cm^{-1} which show a similar decrease after the grafting reaction. The decrease of the signal at 1474 cm^{-1} (Fig. 8d) in the Alkyl-Silik spectrum with respect to the spectrum of as-is resin must be read with the same perspective.

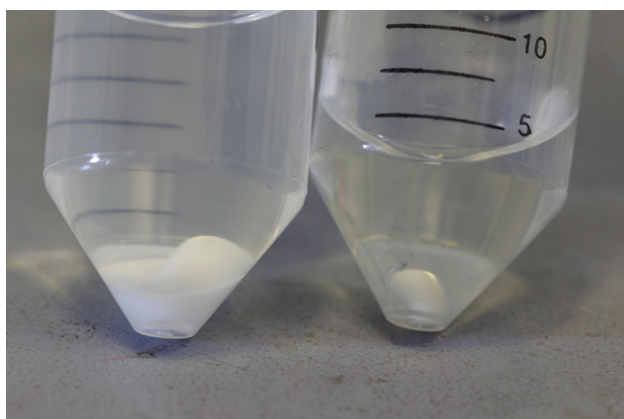


Fig. 6: Separation occurring in fluorinated silane/water systems in high solid system (left); fluorinated silane/water dissolved in acetone (right)

Evaluation of surficial patterning and amphiphilic properties

The wetting hysteresis of each sample was evaluated by the measurement of the advancing, receding, and static contact angles. The measured values of these parameters are reported in Table 5.

Values of advancing contact angle and static contact angles showed the differences in the hydrophobicity of the experimental surfaces. In particular, the sample Alkyl2 featured an average advancing angle of 121° against 114° featured by the F1 sample, and 93° of the Ref.3 sample. The values of static contact angles

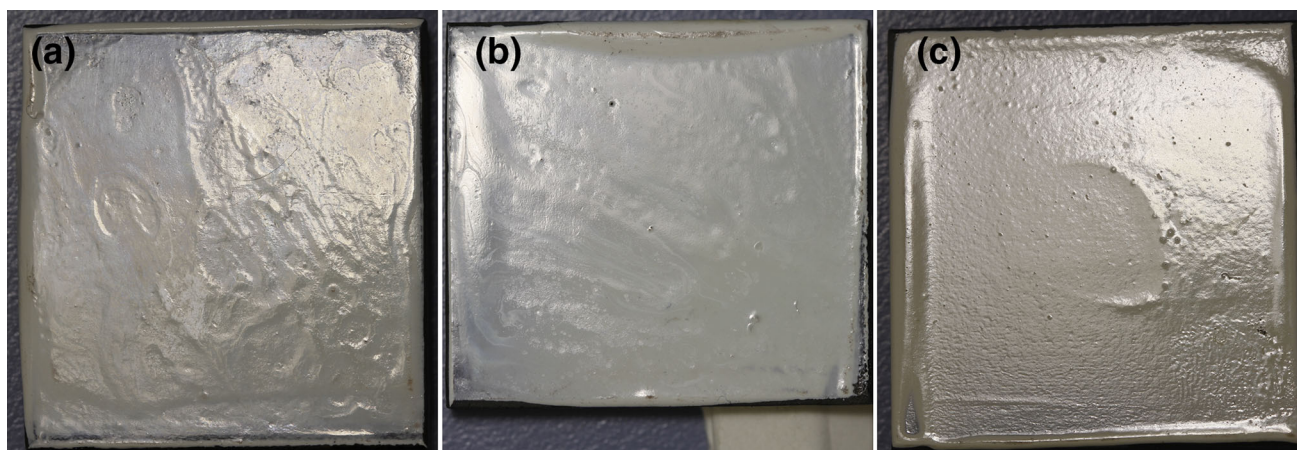


Fig. 7: Segregation phenomenon on samples with a F/PEG ratio of (a) 1:3.4; (b) 1:10; achievement of a correct miscibility among coating components with a F/PEG ratio of 1:15

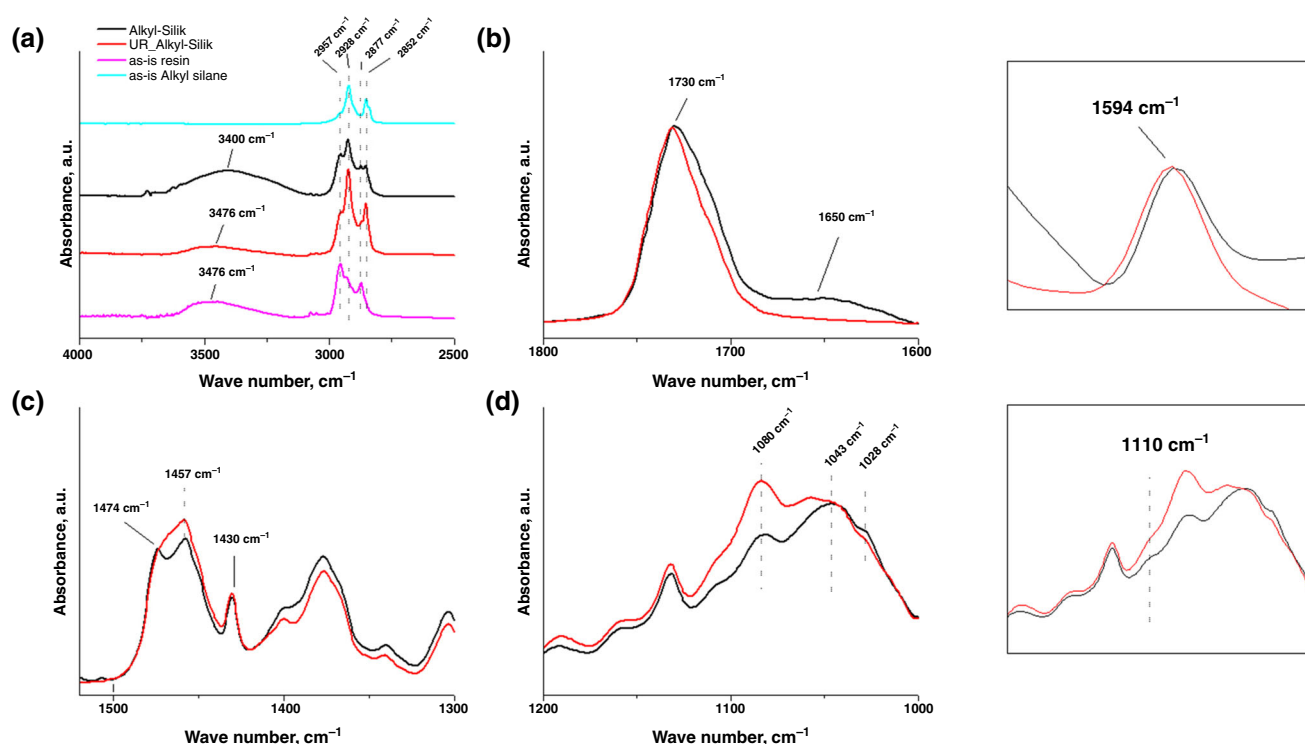


Fig. 8: Images and details of FTIR analysis on grafted resin along different spectral ranges

Table 5: Wetting properties of analyzed samples: values of static, advancing, receding and hysteresis contact angles

Sample ID	Static contact angle	Mean adv. angle	Mean rec. angle	Hysteresis	Adv. angle STD	Rec. angle STD	Hysteresis error
F1	114	114	54	60	4	3	7
Alkyl2	119	121	74	47	5	5	10
Ref.3	92	93	45	48	7	4	11

reflected a similar behavior with 114°, 119°, and 92° measured for the samples F1, Alkyl2, and Ref.3, respectively. The increase in the hydrophobic character of Alkyl2 with respect to F1 was attributed to a series of elements. The main reason for the higher hydrophobicity featured by Alkyl2 was identified with its higher absolute content of hydrophobic silane. The C16-Si silane contained in sample Alkyl2, even if characterized by a higher surface energy,⁵¹ and then characterized by a higher compatibility with the other reactants, was added at higher concentrations (0.01644 mol of C16-Si against 0.00317 mol of FC8-Si). For this reason, it was also more likely the occurrence of a migration of C16-Si toward the outmost surface of Alkyl2 than the migration of FC8-Si in sample F1, even if the C16-Si concentration had not yet reached a critical value such as to induce a visible separation of the phases (cf. “[Assessment of amphiphilic coating formulation](#)” Section). The relationship between the dimensions of

hydrophobic/hydrophilic segregated domains in amphiphilic systems had already been observed in the study of Gan and collaborators.⁵² The authors prepared an amphiphilic coating using a hyper-branched fluoropolymer with diamine-terminated PEG as the crosslinker. The increase in the crosslinker concentration led to a corresponding increase in the amount of PEG-ilate moieties on the surface of the coatings.

The effective occurrence of segregation phenomenon on the surface of sample Alkyl2 was then experimentally evaluated by SEM observation (Fig. 9).

The SE2 detector image of sample F1 showed a modest and regular waviness on the surface (Fig. 9b). This configuration is compatible with a contained, but still present, migration of the fluorinate hydrophobic moiety. Han et al.⁵³ analogously related the onset of a significant roughness on the surface of their coatings with the migration of monomers of acrylic acid and uncrosslinked poly acrylic acid from the bulk of the

binder resin, a polyether sulfone. In the present study, SE2 and RBDS detectors were used in the observation of the F1 and Alkyl2 surfaces. The RBDS analysis of F1 evidenced areas characterized by different average atomic composition, appearing as white cloudy spots in a darker background, compatible with the migration of fluorinate species, the latter having a higher average Z than PEG-ilate chains and the polysiloxane binder (Fig. 9a). The images related to Alkyl2 (Figs. 9c and 9d) evidence a significantly larger phenomenon of segregation occurring on the surface. The images corroborate the hypothesis of massive migration due to the higher concentration of C16-Si, as also suggested by the analysis of the static contact angle and receding angles. Indeed, Alkyl2 featured a receding contact angle of 74° , while F1 showed a behavior indicating a more hydrophilic character, having a receding contact angle of 54° . Among the three, the lowest receding angle was shown by the Ref.3, with a value of 45° , coherently with the values of the advancing contact angle. The higher value of the receding contact angle found with Alkyl2 can also be interpreted as a consequence of a content of hydrophilic PEG-ilate hardener lower than that in sample F1 (0.016 mol against 0.066 mol, respectively). Furthermore, a “hydrophobic moieties content/hydrophilic moieties content” ratio higher in sample Alkyl2 than in F1 can be considered as a possible explanation. The differences

between the values of advancing angle in the F1 and Alkyl2 samples are less pronounced than the differences registered between the two receding angles, leading to wetting hysteresis of 60° , 47° , and 48° for F1, Alkyl2, and Ref.3, respectively. Assuming that the PEG-chains on the hardener feature the same mobility as they are introduced through the addition of the same reagent, the differences in hysteresis values were entirely attributable to the differences in the steric characteristic of the surrounding system that let the PEG-ilate chains rearrange more efficiently in the F1 sample. Indeed, the lower concentration of FC8-Si in formulation F1 led to a more scattered, that is, less packed, distribution of the fluorinate chains given by a lower extension of the segregation phenomenon. In turn, this would give place to a larger space allowing the rearrangement of PEG-ilate chains when in contact with water. The mobility of alkyl chains is further improved by the presence of higher contents of gauche defects. As described in the literature,^{27,28} the number of gauche defects increases as the chain length exceeds 12 C, letting these distortions prevent the thick packing of chains in a rigid, crystalline structure. Consequently, as also supported by the SEM images, it can be deduced that the higher values of hysteresis angles of the fluorinate samples can be attributed to the migration and arrangement of the hydrophilic PEG-ilate segments rather than to the effective achievement of a

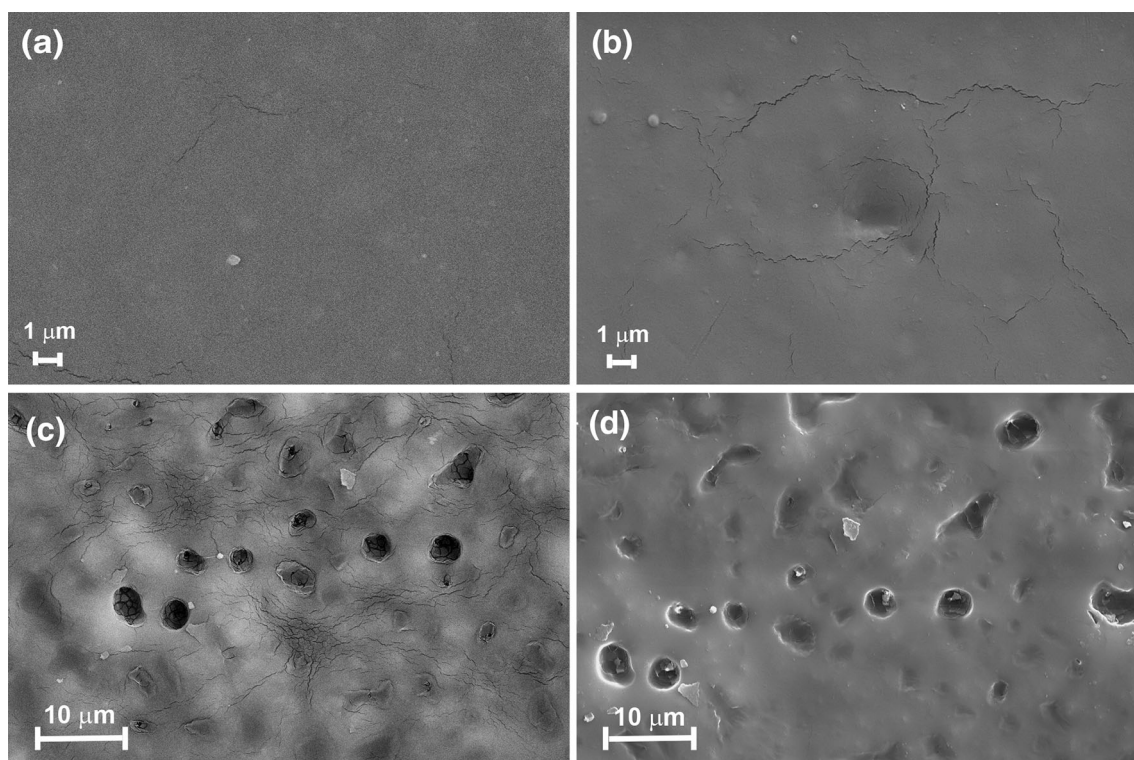


Fig. 9: SEM images of the sample F1 surface acquired through (a) RBDS, and (b) SE2 detector; SEM images of the sample Alkyl2 surface acquired through (c) RBDS, and (d) SE2 detector

finely defined patterning.⁵⁴ The variation of the roughness between sample F1 and sample Alkyl2, even if reported to strongly influence the wettability of a surface,⁵⁵ was, in this context, of secondary importance. Sample F1, characterized by a smooth and regular surface in contrast to sample Alkyl2 which features an irregular patterning on the micrometric scale, displayed a lower value of the advancing contact angle than the Alkyl2 sample, in accordance with the findings of Goli et al.⁵⁶ Those authors stated that the introduction of a patterning on a low energy surface (i.e. originally characterized by a contact angle higher than 90°) would increase the value of the contact angles. If the roughness of the surface was the discriminant element in the determination of a lower wettability of sample Alkyl2 assessed through the measurement of advancing angle, the same element should have affected the receding contact angle in the same way. Nevertheless, despite the roughness of sample Alkyl2, the latter showed a lower value of the receding angle with a consequent increase of the hysteresis angle. This finding is easily explainable, and further emphasizes the role played by the rearrangement of the PEG-ilate segments of the F1 sample.

Antifouling characterization

Egg white test

This section illustrates the results related to the testing conducted for the evaluation of release properties (egg white tests) and the anti-settling activities (testing with marine organisms) of experimental samples. Figure 10 shows images of one representative sample among the three replicates after the brushing step, while Table 6 sets out the marks scored by all the five coated tiles.

As shown by the images, the highest AF/FR efficiency was detained by the samples containing C16-Si which reached a score of “0” (perfectly clean). Despite the lower critical energy featured by FC8-Si contained in the F1 sample, the foul-release character of sample Alkyl2 was by far more efficient in all the tests conducted in this study, in accordance with a lower surficial energy of the overall system.

Antifouling test against mussels

The effective efficiency of the foul-release properties of sample Alkyl2 was further evidenced by the “mussel test”. The results of this test, reported in Table 7, are

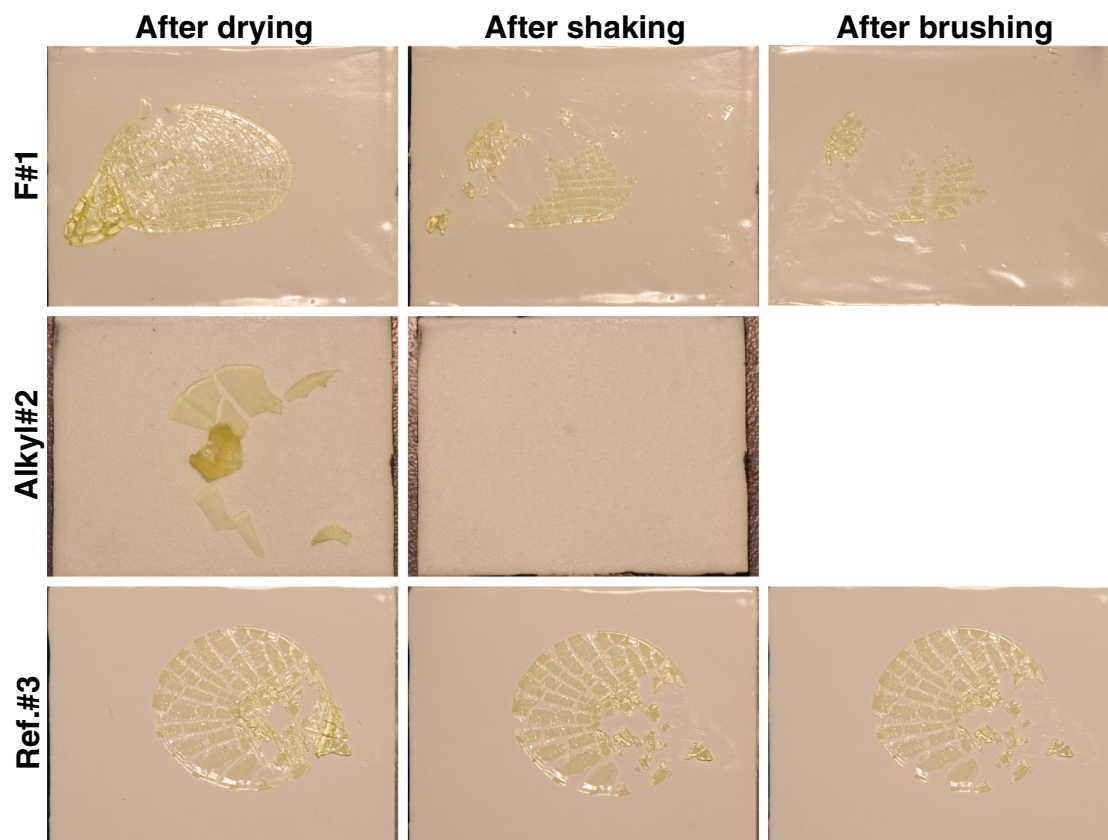


Fig. 10: Visual appearance of the egg white test results; only the most representative replicate was reported for each formulation

dependent on both the adhesion strength and the anti-settlement properties of the surfaces.

The highest achievable grade of AF/FR efficiency in the mussel test, “0”, was only reported by two of the three replicates of sample Alkyl2 (Figs. 11a and 11b). This result corresponds to an easy removal of the byssus plaques attached to the sample (Table 3), emphasizing the effectiveness of foul-release properties of the sample despite the use of a less hydrophobic silane for its preparation. This score is even more significant considering the outstanding adhesive properties of the mussel plaques, since the secreted gluing substance is known as one of the most powerful naturally occurring adhesives. The additional presence of anti-settling properties in sample Alkyl2 is, however, shown by the score achieved by the third replicate. The slab scored “1” indicates the complete absence of any deposited plaque or organism on the tested surface. This result indicates that, in the case in which a mussel is somehow able to reach alternative surfaces (e.g., the plastic bottom of the aquarium), it would prefer to move away rather than adhere to the hostile sample surface. Indeed, a marine organism selects its potential substrate proceeding through several steps of reversible attachment aimed at the evaluation of the substrate until an optimal surface is found for the establishment of a permanent adhesion.⁵⁷ Sample F1 revealed a less efficient antifouling behavior having scored an average mark of 3 (Fig. 11c). The registered behavior was still indicative of the presence of an effective anti-settling action of the examined formulation if compared with the non-antifouling control sample, which constantly scored a mark of 4 for all three replicates (Fig. 11d). Moreover, the attempt to mechanically remove the residual adhesion plaques resulted in the damaging of the most external part of the plaque rather than in its removal with a neat detachment at the interface, in accordance with the adhesion properties of mussel glue.

Table 6: Numerical results of the egg white tests

	Test 1	Test 2	Test 3	Test 4	Test 5	Mean	SD
F1	2	2	3	0	0	1.4	1
Alkyl2	0	0	0	0	0	0	0
Ref.3	2	3	2	3	3	2.6	1

Table 7: Numerical evaluation of the antifouling activity against *Mytilus edulis*

Sample ID	Grade mark of individual repetitions			Grade mark	SD
F1	4	4	2	3	1
Alkyl2	0	0	1	0.3	0.6
Ref.3	4	4	4	4	0

Antifouling test against algae

Finally, the evaluation of antifouling properties against algal organisms was visually and quantitatively conducted. A representative image of the results related to each formulation is shown in Fig. 12, while the quantitative evaluation of the adhered algal species is shown in Table 8.

The highest percentage of residues was found for the white non-antifouling marine paint, with a percentage of the covered area of 6.1%. This result emphasizes the antifouling properties of both F1 and Alkyl2. Indeed, even the less-effective anti-algae formulation, namely F1, features a percentage of area covered by algae of only 2.3%, while sample Alkyl2 showed once again outstanding AF/FR properties with a surface coverage of 0.6%, which confirmed the results obtained in the previous tests.

Discussion on antifouling tests results

Antifouling tests on experimental samples were conducted using the three different testing probes to have an overall view of AF/FR properties of coatings as complete as possible. Indeed, the pertinent literature describes how the antifouling behavior of a surface is strictly related to the specific fouling species. Krishnan¹⁴ reported that low-energy, PDMS-based surfaces were effective in the foul release of algae *Ulva linza*, while single-celled algae *Diatom* spp. was repelled by surfaces with an opposite character. Despite the purpose of the amphiphilic coating system being to join in a single product the advantages of antifouling coatings of two opposite polarities, the chemical nature of the hydrophobic and hydrophilic moieties, and, furthermore, their extension and geometrical arrangement, still influence the overall efficiency of the coating. In this framework, it emerged that the use of alkyl silanes as the replacement of fluorinated silanes for the building of hydrophobic domains does not deplete the efficiency of the coating. Moreover, the spatial arrangement of the moieties in the Alkyl2 sample, as well as its mobility properties, are more favorable for the release of all the analyzed probes. This result could be explained by a series of elements that characterize this formulation. In the first place, surface roughness when present on low energy surfaces aids the prevention of spore and protein deposition, also reducing the adhesion strength of the biofilm, as illustrated by Schumacher et al.^{28,58,59} Those authors

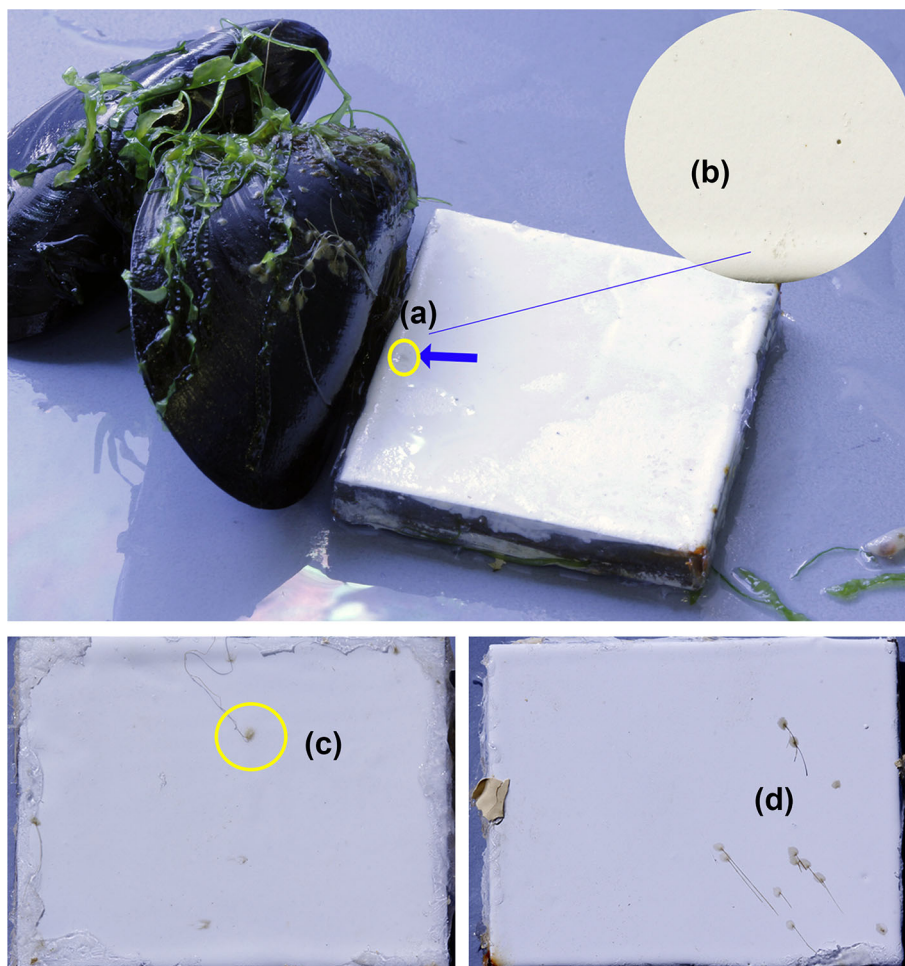


Fig. 11: Samples after the “mussel test”. (a) Sample Alkyl2 before the gentle rubbing shows the presence of only one byssus plaque; (b) sample Alkyl2 after the gentle rubbing shows the complete removal of byssus plaques; (c) sample F1 after the gentle rubbing shows the same aspect as before the rubbing; (d) sample Ref.3 after the gentle rubbing shows the same aspect as before the rubbing



Fig. 12: Algal residues on tested samples after 20 s rinsing

showed that the addition of a regular micro-scale patterning on a PDMS surface would increase both the anti-settling and foul-release properties. The mechanism underneath involves the arrangement of the bio-material on the top of the pillars that decorate the substrate, decreasing the extension of the adhesion interface. In light of this, the Alkyl2 sample, which

visibly features a more pronounced roughness at a micrometer scale compared to F1, has an advantage in protein-release activity, as also evidenced by all the results of the antifouling tests. Differences in the percentage of the surficial area occupied by the low-energy moieties justify the lower adhesion strength of the proteinaceous probes on sample Alkyl2 when

Table 8: Average percentage of covered surface by algal organisms

Sample	Mean pollution %, $\pm$ SD
F1	2.3 ± 1.4
Alkyl2	0.6 ± 0.2
Ref.3	6.1 ± 3.4

compared to sample F1. The latter formulation, even when prepared from fluorinate silanes, provides a larger percentage of the outmost surface dominated by PEG-ilate segments, as evidenced by the values of contact angle hysteresis. PEG-ilate moieties, even if responsible for a more effective anti-settling activity against amphiphilic proteins and glycoproteins,⁶⁰ require higher shear stresses to remove the species eventually adhered on them, since they act as anchor points.⁶¹ This observation can explain the results of the three antifouling tests, since all the testing probes used involve the presence of proteins and glyco-proteins as adhesive agents. The tidy arrangement of hydrophobic chains is a further element that could contribute to the enhancement of the antifouling properties of the Alkyl2 sample.²⁸ The authors prepared a self-assembled monolayer each time using thiols bearing alkyl chains of different lengths and reported a tidy packing for alkyl chains longer than 12 methylene units, justifying this arrangement with the spatial configuration assumed by a thermoplastic polymer's organized crystalline fractions. In contrast, shorter chains, richer in "gauche" defects spread along the chains, usually featured a looser arrangement associated with the amorphous phase of a thermoplastic polymer. These considerations are still valid when a perfluorinated alkyl chain is adopted in place of a purely hydrogenated alkyl segment.⁵⁶ The larger flexibility of fluorinate chains leads to structures that are able to swaddle the adhesive molecules on the molecular level, resulting in a larger contact surface. On its side, the longer alkyl chains on Alkyl2, being organized in a tidier arrangement, result in a more rigid behavior. This configuration more efficiently transmits the shear stresses applied to the surface to the adhesion interface of the biofilm, which, in this way, is more easily removed. Shear stresses applied during the algae and egg white tests gave results in accordance with this mechanism. A last element that led to the achievement of the better performances of sample Alkyl2 was the arrangement in a more efficient geometry of the domains on the Alkyl2 surface. This behavior was also found by Martinelli et al.,¹⁸ who prepared coatings based on PDMS enriched with fluorinate amphiphilic acrylic copolymer, evidencing how, independent from the effective stoichiometric content of fluorine, the AF/FR performances improved with the extent of low-energy domains segregated on the outmost surface. They concluded that the achievement of a fine

patterning played a discriminant role, more than the effective wetting properties of the surface.⁶²

It may be concluded that the introduction of alkyl silanes can achieve, and, under specific conditions, overcome the performances of fluoro-alkyl silanes in the implementation of amphiphilic coatings, both in terms of anti-settling and foul-release properties.

Conclusions

The present work has demonstrated the effectiveness of the use of long alkyl silanes as a green alternative to the use of fluorinate alkyl silanes in the implementation of amphiphilic painting products that provide the functionalization of a commercial polysiloxane resin through an easily implementable reaction pathway.

Indeed, despite the fact that the higher critical energy of the long alkyl silane reactant, with respect to the fluorinate alkyl silane, could lead to a higher critical energy within the surface of a single hydrophobic domain, it also allows the achievement of a series of advantageous conditions. First, the alkyl silane can be added at higher concentrations within the selected binding system, namely a polyester modified, -OH-rich, polysiloxane resin. This higher concentration compensates for the gap between the hydrophobicity of the two silanes and furthermore allows the overcoming of AF/FR effectiveness of fluorine-containing amphiphilic coatings, acting through a series of elements, mainly consisting in consequence of a larger amount of hydrophobic segments on the coating surface:

- A rougher surface that reduces the extension of the contact area with the contaminating species, consequently reducing the energy required to remove the biofilm.
- A more efficient geometrical arrangement of the surface patterning, substantiated by the higher AF/FR properties of Alkyl2 against the adhesion/settling of mussels, egg white and algae.

In addition, the possibility to use silanes bearing longer lateral chains than the ones legally allowed for fluorine-containing composites also resulted in the possibility of the achievement of a semi-crystalline arrangement within the area of hydrophobic domains. The higher rigidity allows a more efficient transmission of the shear stresses applied on the surfaces towards the interface with the biofilm, improving the foul-release properties of the system.

Moreover, the lower hydrophobicity of the long alkyl silane used, which is responsible for the higher concentration achievable of the hydrophobic fraction, also allows the introduction of this reagent without the contextual introduction of TEOS as compatibilizer. This is of utmost importance since it allows the usage of lower contents of PEG-ilate hardener, allowing in

turn the usage of a more appropriate hydrophobic/PEG ratio (nearer to 1:1).

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